Sameer Rathore

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Education

GL Bajaj Institute of Technology and Management **2024 – 2028**
B.Tech – Computer Science & Engineering (AI/ML) | CGPA: 7.9 / 10 *Noida, India*

Relevant Coursework

- Machine Learning
- Deep Learning
- NLP
- Data Structures
- Algorithms
- Database Management
- Theory of Computation
- Computer Networks

Projects

Multi-Agent Code Review System | *LangGraph, ChromaDB, Groq, PyGithub, Pydantic* | GitHub **July 2026**

- Built a LangGraph-based agentic system that reviews GitHub PRs using 3 specialist AI agents (security, style, logic) running **in parallel via the Send API**, each backed by a dedicated RAG pipeline over domain knowledge bases (OWASP Top 10, PEP8, Python anti-patterns).
- Implemented per-agent ChromaDB collections with tuned chunk sizes (350–800 chars) and SentenceTransformers (all-MiniLM-L6-v2); supervisor node merges and ranks findings by severity using Pydantic structured output.
- Ran on real FastAPI PRs – detected SQL injection, missing None checks, and docstring violations across parallel agent runs.

AI Resume Screener | *SBERT, LangGraph, ChromaDB, Pydantic* | GitHub **June 2026**

- Built end-to-end resume screening pipeline using SBERT (all-MiniLM-L6-v2) for semantic similarity scoring between resumes and JDs across 3 job roles (IT, Teacher, Designer) with Fit / Partial Fit / Not Fit classification.
- Curated and manually labelled 94 resume-JD pairs; addressed class imbalance via undersampling and applied data-driven threshold tuning based on score distribution analysis per label class.
- Implemented cosine similarity scoring from scratch using NumPy (dot product / L2 norm) with SBERT embeddings as vector representations.
- Achieved 62 classification accuracy with F1-score of 0.83 on Not Fit class; evaluated using per-class precision, recall, and F1 metrics..

Visual Search Engine | *PyTorch, ResNet, FAISS* | GitHub **Jan 2026**

- Built a visual search engine using **ResNet50** for deep feature extraction (2048-dim embeddings) with ImageNet normalization; indexed vectors into **FAISS** for fast cosine similarity search returning top-k visually similar images.
- Implemented a two-stage pipeline: offline indexing (dataset images → ResNet50 → L2-normalized embeddings → FAISS index) and online query search (user image → embedding → cosine similarity → top-k retrieval).
- Applied `faiss.normalize_L2` to convert embeddings to unit vectors, making cosine similarity equivalent to inner product – improves retrieval accuracy and stability at scale.
- Demonstrated scalable storage design: 2048-dim float32 vectors at 8KB/image, supporting linear scaling from 1K (8MB) to 100K images (800MB) with consistent retrieval performance.

Technical Skills

AI / Agentic: LangGraph, LangChain, RAG Pipelines, Multi-Agent Systems, ChromaDB, FAISS, Pydantic
ML / DL: PyTorch, HuggingFace Transformers, SBERT, SentenceTransformers, ResNet, scikit-learn
LLMs & APIs: Groq (Llama 3.3 70B), Gemini, Structured Output, Prompt Engineering
Tools: GitHub, FastAPI, Streamlit, PyGithub, VS Code, NumPy, Pandas
Languages: Python (primary), C++ (DSA)

Open Source & Achievements

GSSoC'25 – GirlScript Summer of Code **2025**
Open Source Contributor | *View PR* *Remote*

- Active contributor to open-source repositories under GSSoC'25 – collaborated on PRs, code reviews, and feature additions across community projects.

Certification: Intro to Machine Learning – Kaggle